

Sandeep Kumar

Full Stack Developer | Toronto, Ontario, Canada | 647-395-0215 | kkumar.sandeep89@gmail.com

Professional Summary

Highly experienced Senior Full Stack Developer specializing in architecting and deploying scalable Generative AI (GenAI) solutions and microservices, with a core focus on React.js for modern frontends and Node.js (Nest.js) for high-performance backend systems. Proven expertise in integrating AI/LLM technologies (OpenAI API, Gemini, FastMCP, Prompt Engineering, RAG Architectures) to drive product growth and build high-performance, AI-driven applications. Deep understanding of full-stack development, complemented by skills in Python (FastAPI, Flask) and Go. Track record of successful project delivery, leveraging strong DevOps practices (Kubernetes, AWS, CI/CD) and agile methodologies.

Experience

Intuit | Senior Full-stack Developer

Toronto, Canada Area | Nov '24 - May '26

Key Responsibilities:

- Led end-to-end development of scalable microservices using **Node.js (Nest.js)** and **Python (FastAPI)** for high-performance backend systems.
- Hands-on experience in architecting and deploying GenAI-based solutions (LLMs, APIs, agents), leveraging **Model Context Protocol (MCP)** and **RAG Architectures** on AWS/Azure, and implementing prompt engineering.
- Designed and created **React sub-apps** for critical Billing systems, enhancing frontend functionality and user experience.
- Orchestrated technical roadmaps for AI-driven product growth, utilizing JIRA for agile experimentation and ensuring full software development life-cycle with CI/CD practices.
- Maintained security and reliability of microservices by implementing security gateways (**JWT** authentication) and conducting risk assessments.

Projects:

1. AI Analytics: Built predictive models to analyze user behavior, utilizing **RAG-based insights and NLP techniques** to fine-tune sales onboarding flows.
2. Eng Enablement: Streamlined internal event systems using Flask and FastAPI, implementing automated documentation generation via **LLMs**.

Carlton One Engagement | Full-stack Developer (Typescript/Nodejs, React, Python)

Toronto, Canada Area | Jul '19 - May '24

Key Responsibilities:

- Full-stack development of Single Page Applications (SPAs) using **React.js and Vue.js** for the frontend, powered by microservices written in **Node.js (Nest.js and Loopback.io)**.
- Developed and optimized backend microservices using **FastAPI (Python)**, including implementing Redis-based caching to enhance user experience and reduce database stress.
- Implemented **CI/CD pipelines (Bitbucket, PHPUnit, Playwright)** and extended test coverage across various modules (e.g., user registration, cart, checkout).
- Designed and delivered customized client rewards programs (e.g., BMW/MINI) by ingesting data from various sources (SOAP/REST, FTP) to calculate KPI scores and generate leaderboards.
- Contributed to legacy systems using PHP/Laravel while transitioning functionality to microservices and event-based architecture.

Microware Computing & Consulting | Lead Software Developer

Gurgaon, India | Apr '15 - Jun '19

Key Responsibilities:

- Led a team of developers, overseeing the complete software development lifecycle, including database design (MySQL, PostgreSQL, MongoDB) and technical execution.
- Applied working knowledge of **Node.js, React.js, and Python (Flask)** to support project development, alongside core PHP frameworks (Laravel, CodeIgniter).
- Developed crypto buy/sell order functionality with smart contracts (Solidity) and integrated applications with 3rd party APIs using **RESTful and SOAP services (OAuth)**.
- Managed DevOps practices by setting up **GIT flow and CI/CD pipelines using Jenkins**, and utilized AWS API Gateway and Lambda for serverless architecture.

Projects:

1. Technology Development Fund (DRDO): Built a secure LAMP stack platform catering to the full project lifecycle, from registration to fund disbursement, passing a security audit based on OWASP guidelines.
2. Msakhi (Mobile app): Developed cloud web services (LAMP stack) to sync data from Android mobile apps used by health workers in remote areas.
3. BRLF (Bharat Rural Livelihood Foundation): Developed a Government of India fund tracking project in LAMP stack (Laravel), deployed on AWS EC2, including financial progress planning and fund tracking.

Kayako | Linux System Administrator

Jalandhar, India | Jul '13 - Apr '15

Key Responsibilities:

- Install and troubleshoot KAYAKO HELPDESK on a variety of client servers (Linux variants and windows) over a variety of servers (apache, nginx, lighthttpd, etc.)
- Working on AWS instances, Docker instances, VPS, and various control panels.
- Troubleshoot various aspects of email systems, database permission issues, duplicate emails, email PIPING, using PATCH command to patch product files, and tuning PHP and MySQL parameters for performance.

- Writing bash shell scripts and integrating Kayako with new technologies; created docker images. Compile PHP and Nginx on client servers, if they are not supporting package management.
- Complete tasks based on client / process SLA.

About: <http://www.kayako.com/>

Education

CTIEMT Jalandhar | Bachelor's Degree in ECE

May '09 – May '13

Technical Skills

AI & LLMs: OpenAI API, Gemini, FastMCP, Prompt Engineering, RAG Architectures.

Languages: Python (FastAPI, Flask), TypeScript/Node.js (Nest.js), PHP (Laravel, Symfony), Golang, SQL, Bash.

Frontend & Backend: React.js, Vue.js, SASS, MySQL, PostgreSQL, MongoDB, Redis, Kafka, RabbitMQ.

Cloud & DevOps: AWS (Lambda, EC2), Kubernetes, Docker, Jenkins, Git, Jira.

Certifications

Managing Cloud-native Applications with Kubernetes

Neo4j Fundamentals

DevOps Engineering on AWS